(a) Rendered "research-figure" draft



(b) Construction trace in pseudocode

Require: Figure brief B, synthetic data D
Programmatic construction
$S \leftarrow \textsc{InitCanvas}(2048, 1152)$
Draw architecture and arrows in S. ▷ a
Draw bars from D; reserve photo cells. ▷ b,c
Draw learning curves and ablation table. ▷ d,e
Draw schematic contact-state clusters. ▷ f
Asset generation and selection
$L \leftarrow [2\times5,\ 2\times5,\ 3\times3,\ 2\times4]$
for $k = 1, \ldots, 4$ **do**
  $A_k \leftarrow \textsc{GenerateAssets}(B, L_k)$
  Inspect A_k against photo requirements.
  Record provenance; revise the next prompt.
end for
$A^* \leftarrow A_3$ ▷ choice in this run
Composition and verification
Crop and place A^* in the reserved cells.
Overlay labels using code.
$I \leftarrow \textsc{Render}(S)$
Inspect the full image and detail crops.
Add notices for unresolved photo requirements.
$I \leftarrow \textsc{Render}(S)$
$Q \leftarrow \textsc{Review}(I, B)$
Export I as PNG; retain draft status under Q.

Figure 10: **Tracing a "research figure case" to its construction process.** Left: a generated six-panel "research-figure" poster. Right: case-specific pseudocode abstracted from the execution trace (comments a-f refer to panels in the rendered image). The recorded program, asset-generation steps, and reviews make it possible to trace how individual elements were constructed and assessed.

GPT-5.6-Terra each complete 92% of tasks, while only 44% and 48% meet all quality thresholds. These observations suggest that general benchmark performance captures only part of the demands of visual program generation, which also requires translating spatial and appearance constraints into executable representations and using rendered feedback to guide revision.

What can the construction process reveal beyond the final image? MaLiang-Harness makes the construction of a visual result inspectable through its executable program, asset provenance, and revision history. Fig. 10 pairs a "six-panel research illustration" of an academic paper with pseudocode summarizing its recorded construction process. GPT-6-Astra uses code to specify the layout, architecture diagram, labels, and plots from user-supplied synthetic data, while generating photographic-style assets for the manipulation examples. The trace records four asset-generation attempts with different contact-sheet layouts. The final composition uses crops from the third attempt, with code controlling their placement, scaling, and labels. The construction record distinguishes elements drawn from explicit data and instructions from those assembled using generated imagery. **This distinction also helps interpret the final review**: the numerical plots and layout pass, whereas the photographic examples remain unsatisfactory, leaving the output only as a draft. By linking construction steps to revision-specific assessments, the harness reveals how individual elements were produced and which requirements remain unresolved beyond what the final image alone shows.

When does refinement stall, and when should it stop? The harness faces three potential failure modes: **reasoning stagnation**, where planning does not advance the artwork; **semantic livelock**, where repeated edits leave requirements unresolved; and **infinite agentic loops**, where planning and tool use repeat without termination. Refinement stalls when continued activity produces no meaningful visual improvement. PEG preserves the working state, while TGP and REV support inspection, comparison, and recovery, but these mechanisms do not guarantee effective corrections. In Fig. 8, further inspection yields no corrective edit before the token budget is exhausted. The harness therefore separates termination from completion: model-call, tool-call, time, and token limits bound execution, while successful finalization is required to mark a task completed.

5 Conclusion

We presented MaLiang-Harness, a unified framework for programmable image and video generation motivated by the Program-to-Visual (P2V) gap between program-level correctness and visual re-